The Psychology of Data Visualization

Fall 2026 ▪ MWF 3:35-4:25p

2026-07-16

Spring 2025 ▪ TTh 9:05-10:25 AM



Figure 1: ChatGPT/Dall-E-3 model response to 'an image of "a picture paints a thousand words"' on 2024-09-19

Themes

If a picture's worth a thousand words (Wikipedia contributors 2024a), what exactly does it say? This course will focus on the psychology of data visualization—how to read, critique, and generate meaningful figures that inform but don't mislead. We'll take inspiration from recognized classic figures and unpack what makes them exemplary. We'll critique figures that

deliberately or inadvertently mislead or confuse. We'll learn what 150 years of vision science, the foundation of experimental psychology, has to say about data visualization. And we'll learn how to build our own reproducible figures using Python, R, and JavaScript. No prior programming experience is required.

Instructor

Rick O. Gilmore, Ph.D.
Professor of Psychology
rog1 AT-SIGN psu PERIOD edu

Office hours

Monday & Friday, 12:00p - 1:00p in 114 Moore or by appointment: https://doodle.com/mm/rickgilmore/book-a-time

Lab web site: https://gilmore-lab.github.io

Teaching Assistant

Kalvin Garrah
Graduate Student in Social Psychology
kmg7035 AT-SIGN psu PERIOD edu

Office hours

Tuesday, 12:00p - 2:00p, in 473 Moore or by appointment (email).

Meeting time & location

Monday, Wednesday, & Friday, 3:35 pm - 4:25 pm
216 Osmond Lab

Canvas site

Most of the course content will be found on this site.

However, we will use Canvas to track attendance, submit and grade assignments, and provide ongoing summaries about grade point totals. The Canvas site may be found here:

https://psu.instructure.com/courses/2484925

Course structure

This is a discussion-focused course. On most days we will discuss readings assigned prior to class. On many days, we will work together or individually on the assigned exercises, the final project, or another assignment. Please remember to bring your personal laptop computer. If you do not have a personal laptop, contact Dr. Gilmore so that we can make arrangements for you to participate fully in the course activities.

Week 01

Surveying the landscape

Mon, Aug 24

Course introduction

- *Watch*:
  - PBSoffbook (2013)
- Slides

Wed, Aug 26

The semiotics of data visualization

- *Read*:
  - Cairo (2013), Chapter 1-2, PDF on Canvas
- *Assigned*: Exercise 01 assigned
- Slides
- *Homework*:
  - Install R and RStudio Desktop

        * *Tutorial*
    - (Optional) Install Quarto

Fri, Aug 28

Work Session: Introducing R & RStudio

- *Complete*: R and RStudio Desktop installation and setup
- *Read*: Get Started from the RStudio User Guide.
- *Skim*:
    - https://quarto.org/docs/get-started/hello/jupyter.html[1]
    - https://education.rstudio.com/learn/beginner/
- Slides

Week 02

Who visualizes data and why

Mon, Aug 31

Visualization in government & business

- Slides

Wed, Sep 02

Visualization in art, sports, and journalism

- *Read*
    - Gelman and Unwin (2013) (recommended), follow citation link to online version or PDF on Canvas.
- *Assigned*: Exercise 02
- Slides
- Results from Exercise 01

[1]We won't use Python in this course, but the basic ideas apply to Python and R.

Fri, Sep 04

Work Session: Exercise 02 and Quarto

- *Due* Exercise 01 | *Canvas dropbox*
- Slides

Week 03

Understanding figures

Mon, Sep 07

NO CLASS: Labor Day

Wed, Sep 09

Making (sense of) data

- *Read*:
  - Stevens (1946) | PDF on Canvas
  - Wikipedia contributors (2024b) (Optional)
  - Tourangeau et al. (2012) | PDF on Canvas. (Optional)
- Slides

Fri, Sep 11

Work Session: Making data with R

- Slides

Week 04

The anatomy of figures

Mon, Sep 14

Figure types

- *Explore* (pick at least two):
  - Ribecca (n.d.)
  - Hammond (2024)
  - Discovery (2024)
  - Atlassian (n.d.)
- *Watch*:
  - Channel (2024)
- Assigned: Exercise-03
- Slides

Wed, Sep 16

More figure types and figure components

- Read: Tutorial: Making data
- Slides

Fri, Sep 18

Work Session: Intro to ggplot

- Slides
- *ggplot2* tutorial
- Continue work on Exercise 02 and Exercise 03.

Week 05

Perceptual foundations

Mon, Sep 21

NO CLASS

Wed, Sep 23

From stimulus to sensation

- *Read*:
  - Cairo (2013), Chapter 5, PDF on Canvas
- Slides

Fri, Sep 25

Work Session: More ggplot

- *Due* Exercise 02 | *Canvas dropbox* |
- Tutorial

Week 06

Visual perception and cognition

Mon, Sep 28

From sensation to perception

- *Read*:
  - Cairo (2013), Chapter 6, PDF on Canvas
  - Few (2004), Chapter 5, PDF on Canvas
- *Optional*:
  - Cleveland and McGill (1984)

- Slides

Wed, Sep 30

From cognition to understanding

- *Read*:
  - Franconeri et al. (2021), pp. 109-122, PDF on Canvas
  - Cairo (2013), Chapter 7, PDF on Canvas
- Assigned: Final Project proposal
- Slides

Fri, Oct 02

Work Session: Visualizing sports data

- *Due* Exercise-03 | *Canvas dropbox*](https://psu.instructure.com/courses/2484925/assignments/18448437) |
- Tutorial
- Slides

Week 07

Mon, Oct 05

Work session: Making figures on your own I

- Slides

Wed, Oct 07

Work session: Making figures on your own II

- Slides

Fri, Oct 09

Work Session

- Slides

Week 08

Mon, Oct 12

Critiquing figures

- *Read*:
  - Tufte (2001), Chapter 1, PDF on Canvas
  - Tufte (2001), Chapter 2, PDF on Canvas
- Slides

Wed, Oct 14

Storytelling with data

- *Read*:
  - Knaflic (2015), pp. ix-33, PDF on Canvas
  - Knaflic (2015), pp. 33-69, PDF on Canvas
- *Skim*: Woodside et al. (2008), PDF on Canvas
- Assigned: Exercise-05
- Slides

Fri, Oct 16

Work Session: Making data tidy

- *Due* Exercise-04 | *Canvas dropbox* |
- *Due* Final Project proposal | *Canvas dropbox*
- Tutorial

Week 09

Visualizing spatial data

Mon, Oct 19

Map-making

- Read:
  - Wilke (2019), Chapter 15, PDF on Canvas
- Slides

Wed, Oct 21

- Slides

Fri, Oct 23

Work Session: TidyTuesday

- Slides

Week 10

Turning the table(s)

Mon, Oct 26

Tables in Base R

- *Due* Exercise-05 | *Canvas dropbox*
- Slides

Wed, Oct 28

`gt` and `knitr::kable`

- Slides

Fri, Oct 30

Work Session: Tables

- Slides

Week 11

Data viz in policy & politics

Mon, Nov 02

- Slides

Wed, Nov 04

- Slides

Fri, Nov 06

Week 12

Mon, Nov 09

Wed, Nov 11

Fri, Nov 13

Work Session: More TidyTuesday

- Slides

Week 13

Data viz and quantitative inference

Mon, Nov 16

Synthesize before you analyze

- Slides
- *Due* Exercise-06

Wed, Nov 18

Sample size, effect size, statistical power

- Slides

Fri, Nov 20

Work Session: Final projects

Thanksgiving Break

NO CLASS

Week 14

Mon, Nov 30

Making data dashboards and apps

- Explore:
  - Kohler (2024)
- Slides

Wed, Dec 02

Work Session: Final projects

Fri, Dec 04

Work Session: Final projects

Week 15

Mon, Dec 07

Final Project Presentations

Wed, Dec 09

Final Project Presentations

Fri, Dec 11

Final Project Presentations

Finals Week

Wed, Dec 16

- *Due* Final project write-up | *Canvas dropbox*

Elements

Component	Description	Points
Attendance	You will receive 1 point for each class you attend up to a maximum of 40.	40
Exercises	There will be six (6) exercises that we will work on. Each exercise is worth 10 points. You are required to submit four (4) toward your final grade.	40
Final project	You will complete a final project, either on your own, or with a small group of 3 or fewer. Your final project is worth 40 points.	40
	TOTAL POINTS POSSIBLE	120
Extra Credit	If you submit more than four (4) exercise write-ups, you may earn up to 10 extra credit points for these exercises.	

Grading Scheme

Percent	Points	Grade
94+	113+	A
90-93	108-112	A-
87-89	104-107	B+
84-86	100-103	B
80-83	96-100	B-
77-79	92-95	C+
70-76	84-91	C
60-69	72-83	D
<59	<=71	F

This page summarizes some of the key deadlines in the course.

Date	What's due/happening
2026-09-04	Exercise 01
2026-09-25	Exercise 02
2026-10-02	Exercise 03
2026-10-16	Exercise 04
2026-10-16	Final project proposal
2025-10-26	Exercise 05
2026-11-16	Exercise 06
2026-12-16	Final project writeup due

Figure Figure 2 below shows one way to visualize these deadlines. We'll examine it and attempt to improve it at some point in the class.

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

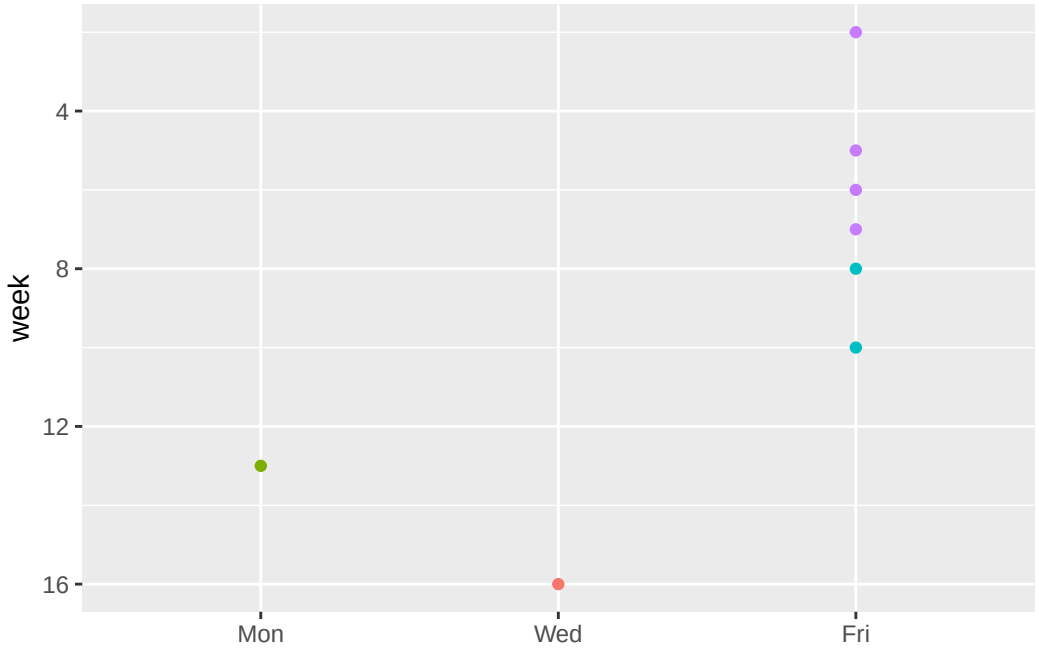


Figure 2: Visualization of assignment deadlines in PSYCH 490.007 Fall 2026 by week of the semester, day of the week, and month.

Academic Integrity

Students with questions about academic integrity should visit http://www.la.psu.edu/current-students/undergraduate-students/education/academic-integrity.

Penn State defines academic integrity as the pursuit of scholarly activity in an open, honest and responsible manner. All students should act with personal integrity, respect others dignity, rights and property, and help create and maintain an environment in which all can succeed through the fruits of their efforts (Faculty Senate Policy 49-20). Sanctions for academic misconduct can include a grade of F for the course as well as other penalties.

Unless you are told otherwise, you must complete all course work entirely on your own, using only sources that have been permitted by your instructor, and you may not assist other students with papers, quizzes, exams, or other assessments. If I allow you to use ideas, images, or word phrases created by another person (e.g., from Course Hero or Chegg) or by generative technology, such as ChatGPT, you must identify their source.

When you complete assignments, remember the **ABC**s to avoid plagiarism: **A**lways place copied information within quotation marks, include information about the quoted or paraphrased source in a **B**ibliography, and **C**ite the source in the body (in the text) of your paper immediately

after the quoted or paraphrased information. When in doubt, cite in the text and include the source in a bibliography.

Students with questions about academic integrity should ask me or the TA before submitting work.

Students facing allegations of academic misconduct may not drop/withdraw from the affected course unless they are cleared of wrongdoing (see G-9: Academic Integrity). Attempted drops will be prevented or reversed, and students will be expected to complete course work and meet course deadlines. Students who are found responsible for academic integrity violations face academic outcomes, which can be severe, and put themselves at jeopardy for other outcomes which may include ineligibility for Dean's List, pass/fail elections, and grade forgiveness. Students may also face consequences from their home/major program and/or The Schreyer Honors College.

Use of Artificial Intelligence (AI)

You may use AI tools to assist with your work, but you must properly cite any sources used. This includes AI-generated content, such as text, images, data, or code.

Always ensure that your work reflects your own understanding and that you are not relying solely on AI tools to complete assignments.

If you do make use of AI, please cite the AI tool used and provide a brief description of how it was used in your work. Please also save a transcript of your interaction with the AI tool and submit that as an appendix to your assignment.

See https://integrity.psu.edu/ai for more details, and when in doubt, please ask the instructor or TA.

Absences or late assignments

Absence from class

Your absence from class may be excused under unusual circumstances such as (a) an interview for graduate school or a job, (b) illness, (c) religious observance, (d) the death of a family member, or (e) any other event recognized by the university as a valid excuse for absence from class.

If you must miss class, you must contact the instructor and the TA **in advance**.

Up to three (3) excused absences will be permitted.

Late exercises

Exercises submitted after the published deadlines will not be eligible for full credit unless the instructor has given specific permission.

Late final projects

Final projects submitted after the published deadline will not be eligible for full credit unless the instructor has given specific permission.

Accommodation for persons with disabilities

Penn State welcomes students with disabilities into the University's educational programs. Please refer to the information provided by Student Disability Resources (SDR) at http://equity.psu.edu/student-disability-resources/ for information about the procedures required to obtain reasonable accommodations in this course. Students should discussSDR-approved accommodations with their instructor as early in the semester as possible, even if they have taken another course with the instructor. Please note: students are not required to provide their instructor with information about the nature of their condition.

Penn State students are also welcome to contact other units for assistance with personal concerns that interfere with academic progress, including: Counseling and Psychological Services (CAPS; http://studentaffairs.psu.edu/counseling/), the Office of Student Affairs (http://studentaffairs.psu.edu/), Career Services (http://studentaffairs.psu.edu/career/), the Center for Women Students (http://studentaffairs.psu.edu/womenscenter/), the LGBTQA Student Resource Center (http://studentaffairs.psu.edu/lgbtqa/), the Office of Sexual Misconduct Prevention and Response (http://titleix.psu.edu/), Penn State Educational Equity (http://equity.psu.edu/), the Multicultural Resource Center (http://equity.psu.edu/mrc), and University Health Services (http://studentaffairs.psu.edu/health/).

Nondiscrimination Statement

The Pennsylvania State University is committed to equal access to programs, facilities, admission and employment for all persons. It is the policy of the University to maintain an environment free of harassment and free of discrimination against any person because of age, race, color, ancestry, national origin, religion, creed, service in the uniformed services (as defined in state and federal law), veteran status, sex, sexual orientation, marital or family status, pregnancy, pregnancy-related conditions, physical or mental disability, gender, perceived gender, gender identity, genetic information or political ideas.
Discriminatory conduct and harassment, as well as sexual misconduct and relationship violence, violates the dignity of individuals, impedes the realization of the University's educational mission, and will not be tolerated.

Direct all inquiries regarding the nondiscrimination policy to:

Suzanne C. Adair Associate Vice President for Equal Opportunity and Access Office of Equal Opportunity and Access The Pennsylvania State University 328 Boucke Building University Park, PA 16802-5901 Email: sca917@psu.edu Tel (814) 863-0471

Diversity Statement

This classroom is a place where you will be treated with respect. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class.
Penn State is committed to creating an educational environment which is free from intolerance directed toward individuals or groups and strives to create and maintain an environment that fosters respect for others as stated in Policy AD29 Statement on Intolerance.

Mandated Reporting Statement

Penn State's policies require me, as a faculty member, to share information about incidents of sex-based discrimination and harassment (discrimination, harassment, sexual harassment, sexual misconduct, dating violence, domestic violence, stalking, and retaliation) with Penn State's Title IX coordinator or deputy coordinators, regardless of whether the incidents are stated to me in person or shared by students as part of their coursework. For more information regarding the University's policies and procedures for responding to reports of sexual or gender-based harassment or misconduct, please visit http://titleix.psu.edu.

Additionally, I am required to make a report on any reasonable suspicion of child abuse in accordance with the Pennsylvania Child Protective Services Law.

Zoom

At some point in the semester, I may decide to use Zoom to allow students who are unable to attend class in person to participate.

While you are on Zoom, keep in mind that this is a classroom environment and others should be treated with respect. Please keep your microphone muted unless you want to ask a question or interact with someone. If your microphone is not muted, the entire class will be able to hear what is going on in your environment. As an instructor, I personally like to see people's faces. As a participant, I am more involved when I have my camera on. I realize, however, that there are many reasons why you might not want to turn on your camera such as poor internet connection, joining via phone, or other privacy concerns. It is your choice as to whether you would like to have the camera on or not.

Values

Penn State Principles

The Pennsylvania State University is a community dedicated to personal and academic excellence. The Penn State Principles were developed to embody the values that we hope our students, faculty, staff, administration, and alumni possess. At the same time, the University is strongly committed to freedom of expression. Consequently, these Principles do not constitute University policy and are not intended to interfere in any way with an individual's academic or personal freedoms. We hope, however, that individuals will voluntarily endorse these common principles, thereby contributing to the traditions and scholarly heritage left by those who preceded them, and will thus leave Penn State a better place for those who follow.

I will respect the dignity of all individuals within the Penn State community. The University is committed to creating and maintaining an educational environment that respects the right of all individuals to participate fully in the community. Actions motivated by hate, prejudice, or intolerance violate this principle. I will not engage in any behaviors that compromise or demean the dignity of individuals or groups, including intimidation, stalking, harassment, discrimination, taunting, ridiculing, insulting, or acts of violence. I will demonstrate respect for others by striving to learn from differences between people, ideas, and opinions and by avoiding behaviors that inhibit the ability of other community members to feel safe or welcome as they pursue their academic goals.

I will practice academic integrity. Academic integrity is a basic guiding principle for all academic activity at Penn State University, allowing the pursuit of scholarly activity in an open, honest, and responsible manner. In accordance with the University Code of Conduct, I will practice integrity in regard to all academic assignments. I will not engage in or tolerate acts of falsification, misrepresentation or deception because such acts of dishonesty violate the fundamental ethical principles of the University community and compromise the worth of work completed by others.

I will demonstrate social and personal responsibility. The University is a community that promotes learning; any behaviors that are inconsistent with that goal are unacceptable. Irresponsible behaviors, including alcohol or drug abuse and the use of violence against people or property, undermine the educational climate by threatening the physical and mental health of members of the community. I will exercise personal responsibility for my actions and I will make sure that my actions do not interfere with the academic and social environment of the University. I will maintain a high standard of behavior by adhering to the Code of Conduct and respecting the rights of others.

I will be responsible for my own academic progress and agree to comply with all University policies. The University allows students to identify and achieve their academic goals by providing the information needed to plan the chosen program of study and the necessary educational opportunities, but students assume final responsibility for course

scheduling, program planning, and the successful completion of graduation requirements. I will be responsible for seeking the academic and career information needed to meet my educational goals by becoming knowledgeable about the relevant policies, procedures, and rules of the University and academic program, by consulting and meeting with my adviser, and by successfully completing all of the requirements for graduation.

Penn State Values

Integrity: We act with integrity and honesty in accordance with the highest academic, professional, and ethical standards.

Respect: We respect and honor the dignity of each person, embrace civil discourse, and foster a diverse and inclusive community.

Responsibility: We act responsibly, and we are accountable for our decisions, actions, and their consequences.

Discovery: We seek and create new knowledge and understanding, and foster creativity and innovation, for the benefit of our communities, society, and the environment.

Excellence: We strive for excellence in all our endeavors as individuals, an institution, and a leader in higher education.

Community: We work together for the betterment of our University, the communities we serve, and the world.

References

Atlassian. n.d. "Essential Chart Types for Data Visualization." Accessed January 24, 2025. https://www.atlassian.com/data/charts/essential-chart-types-for-data-visualization.

Cairo, Alberto. 2013. *The Functional Art: An Introduction to Information Graphics and Visualization*. New Riders Publishing.

Channel, Mconsultingprep Official. 2024. *Every CHART Types Explained in 12 Minutes*. YouTube. https://www.youtube.com/watch?v=cuMwWvtNmJ0.

Cleveland, William S, and Robert McGill. 1984. "Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods." *Journal of the American Statistical Association* 79 (September): 531–54. https://doi.org/10.1080/01621459.1984.10478080.

Discovery, Jmp Statistical. 2024. "Types of Graphs." January 4. https://www.jmp.com/en_us/statistics-knowledge-portal/exploratory-data-analysis/types-of-graphs.html.

Few, Stephen. 2004. *Show Me the Numbers: Designing Tables and Graphs to Enlighten*. Analytics Press.

Franconeri, Steven L, Lace M Padilla, Priti Shah, Jeffrey M Zacks, and Jessica Hullman. 2021. "The Science of Visual Data Communication: What Works." *Psychological Science in the Public Interest: A Journal of the American Psychological Society* 22 (3): 110–61. https://doi.org/10.1177/15291006211051956.

Gelman, Andrew, and Antony Unwin. 2013. "Infovis and Statistical Graphics: Different Goals, Different Looks." *Journal of Computational and Graphical Statistics: A Joint Publication of American Statistical Association, Institute of Mathematical Statistics, Interface Foundation of North America* 22 (January): 2–28. https://doi.org/10.1080/10618600.2012.761137.

Hammond, Tony. 2024. "20 Types of Charts and Graphs for Data Visualization." November 26. https://www.thoughtspot.com/data-trends/data-visualization/types-of-charts-graphs.

Knaflic, Cole Nussbaumer. 2015. *Storytelling with Data*. John Wiley & Sons.

Kohler, Tanner. 2024. "Psychology for UX: Study Guide." January 10. https://www.nngroup.com/articles/psychology-study-guide/.

PBSoffbook. 2013. *The Art of Data Visualization | Off Book | PBS Digital Studios*. YouTube. https://www.youtube.com/watch?v=AdSZJzb-aX8&t=3s.

Ribecca, Severino. n.d. "The Data Visualisation Catalogue." Accessed January 24, 2025. https://datavizcatalogue.com/index.html.

Stevens, S S. 1946. "On the Theory of Scales of Measurement." *Science (New York, N.Y.)* 103 (June): 677–80. https://doi.org/10.1126/science.103.2684.677.

Tourangeau, Roger, Lance J Rips, and Kenneth Rasinski. 2012. *The Psychology of Survey Response*. Cambridge University Press. https://doi.org/10.1017/cbo9780511819322.

Tufte, Edward R. 2001. *The Visual Display of Quantitative Information*. Graphics Pr.

Wikipedia contributors. 2024a. *A Picture Is Worth a Thousand Words*. Https://en.wikipedia.org/wiki/A_picture_is_worth_a_thousand_words; Wikimedia Foundation, Inc.

Wikipedia contributors. 2024b. "Statistical Data Type." Wikimedia Foundation, Inc., November 23. https://en.wikipedia.org/wiki/Statistical_data_type.

Wilke, Claus O. 2019. *Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures*. O'Reilly Media. https://www.amazon.com/Fundamentals-Data-

Visualization-Informative-Compelling/dp/1492031089.

Woodside, Arch G, Suresh Sood, and Kenneth E Miller. 2008. "When Consumers and Brands Talk: Storytelling Theory and Research in Psychology and Marketing." *Psychology & Marketing* 25 (February): 97–145. https://doi.org/10.1002/mar.20203.